When Less is More: Optimal Scope of Public Communication in Networks

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Motivation: Scope Matters

- Public authorities communicate to **large, diverse** populations; public communication is **powerful** to manage expectations and actions.
  - Draghi (2012), *"whatever it takes"* — markets repriced rapidly.
- Recipients are not crowds of *isolated* listeners.
  - E.g., supply chains, interbank markets, social media, etc.
- Same information, different **audience**, opposite result:
  - **Northern Rock (2007):** the public announcement of liquidity support made depositors think the bank was in trouble $\Rightarrow$ a bank run
  - **Ebola epidemic (2014-16):** health messages was directed first to religious leaders $\Rightarrow$ safe-burial intentions nearly doubled

Research question: Broadcast or Target?

- In practice, authorities actively choose the **scope** of communication — broadcasting, yet also *targeting* to particular audiences:
  - **Central banking:** public statements to the market, selective briefings to dealers & journalists (Meyer, 2004; Cieslak et al., 2019).
  - **Social programs:** nationwide advisories, campaigns routed through community leaders (Kim et al., 2015; Banerjee et al., 2019).
- **Key question:** Should a public authority disclose information to a *broad audience*, or only to a *targeted few*?

This Paper's Work and Main Results

- We develop a model of public communication in networks, where:
  - The network captures who follows whom.
  - The planner chooses whom to inform (*scope*).
- **Finding 1: More is typically optimal.**
  - A benevolent planner often prefers broadcasting information to a broad audience across a wide range of network structures.
- **Finding 2: Less can be more.**
  - A smaller scope can improve welfare — but only when some agents are both *central* and *influential*.
- **Finding 3: Targeting the few.**
  - Key *recipient*: a simple *degree* rule; Key *group*: NP-hard in general.

Related Literature and Contribution

- **Targeting in network games** (Ballester et al., 2006; Zhou and Chen, 2015; Demange, 2017; Galeotti et al., 2020; Sun et al., 2023).

  + *target information, not actions or links.*

- **Dispersed information in networks** (Calvó-Armengol and de Martí, 2009; Myatt and Wallace, 2019; Leister, 2020).

  + *endogenous information design: who is informed.*

- **Social value of public information** (Morris and Shin, 2002; Angeletos and Pavan, 2007; Ui and Yoshizawa, 2015; Murayama, 2025).

  + *targeted public information; network positions matter.*

The Model

Payoff Structure

$$u_i(\mathbf{a}, \theta; \mathbf{G}) \;=\; -\underbrace{(a_i - \theta)^2}_{\textit{adaptation}} \;-\; \sum_{j \neq i} g_{ij} \underbrace{(a_i - a_j)^2}_{\textit{coordination}}. \tag{1}$$

- Agents $i \in N = \{1, \ldots, n\}$ choose an action $a_i \in \mathbb{R}$.
- An unknown fundamental $\theta \in \mathbb{R}$.
- Primitive network $\mathbf{G} = (g_{ij})$ is *directed* and *weighted*:
  - **Direction**: $i \to j$ — i follows j, i.e., a *listening network*;
  - **Weight**: $g_{ij} \geq 0$ — i's coordination motive with j (*attentiveness*).

The Planner's Problem

- The planner is benevolent, which maximizes utilitarian welfare:

$$W(\mathbf{a}, \theta; \mathbf{G}) = \sum_{i \in N} u_i(\mathbf{a}, \theta; \mathbf{G}). \tag{2}$$

- Network structure affects welfare in two ways:
  - **Direct**: the coordination motives g_{ij};
  - **Indirect**: the equilibrium actions $\mathbf{a}$.

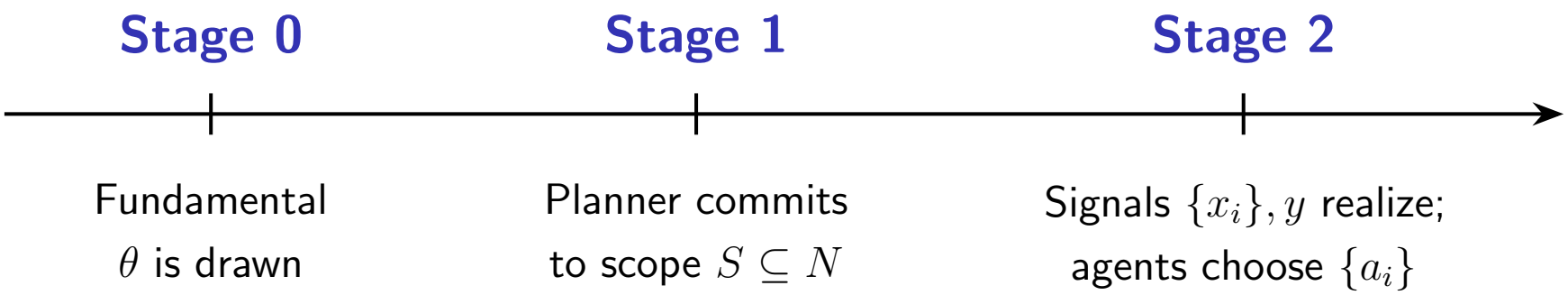
- Public communication is modeled by releasing a "public" signal to the audience (e.g., public announcement).

- **Scope:** the planner commits to a **target set** of recipients $S \subseteq N$ *before* releasing the signal.

Information Structure

- Improper uniform prior on the fundamental θ over $\mathbb{R}$.
- Private signal: $x_i = \theta + \varepsilon_i, \;\; \varepsilon_i \sim \mathcal{N}(0, \tau_x^{-1})$.
- (*Semi-*)Public signal: $y = \theta + \varepsilon_0, \;\; \varepsilon_0 \sim \mathcal{N}(0, \tau_y^{-1})$; only S observes it.
- Assume θ and the noise terms are mutually independent.
- $\gamma \equiv \frac{\tau_y}{\tau_x + \tau_y} \in (0, 1)$ is the relative precision of the public signal ($\gamma \to 0$ fully opaque, $\gamma \to 1$ fully transparent).

Timeline


Stage 0
Stage 1
Stage 2
Fundamental
θ is drawn
Planner commits
to scope $S \subseteq N$
Signals $\{x_i\}, y$ realize;
agents choose $\{a_i\}$

Equilibrium Characterization and Scope Design

How Agents Use Information in Equilibrium

Theorem 1

Given a target set $S \subseteq N$, there is a unique linear equilibrium:

(i) *nonrecipients ($i \notin S$) follow their private signal, $a_i^* = x_i$;*

(ii) *recipients ($i \in S$) blend the two signals,*

$$a_i^* = \left(1 - \gamma\, b_i^S\right) x_i + \gamma\, b_i^S\, y,$$

where the **Katz–Bonacich centrality** *of i within S is*

$$b_i^S = \left[(\mathbf{I} - (1-\gamma)\widetilde{\mathbf{G}}_S)^{-1}\mathbf{1}\right]_i = \sum_{k=0}^{\infty}(1-\gamma)^k \left[\widetilde{\mathbf{G}}_S^{\,k}\mathbf{1}\right]_i.$$

- Katz–Bonacich centrality measures the amplification of public-information use through strategic coordination.
- Recipients over-react to the public signal, and the over-reaction is proportional to his centrality.

Designing the Scope

Theorem 2

Choosing the optimal scope is equivalent to maximizing a closed-form index:

$$\max_{S \subseteq N} \mathcal{T}(S) \equiv \sum_{i \in S} b_i^S \left[\left(1 + d_i^{\text{out}} - d_i^{\text{in}}\right) b_i^S + 2\, d_i^{\text{in}} \right]. \tag{3}$$

- Three network statistics determine the optimal scope:
  - **Out-degree** $d_i^{\text{out}} \equiv \sum_{j \neq i} g_{ij}$ — coordination motive;
  - **In-degree** $d_i^{\text{in}} \equiv \sum_{j \neq i} g_{ji}$ — social influence;
  - **Katz–Bonacich centrality** b_i^S — noise amplification.

- **Trade-off:** influential and central agents over-react to the public signal and propagate its noise. Sufficiently noisy ($\gamma \to 0$) $\Rightarrow$ a broader audience is not optimal.

Broadcast or Target?

When Is Broadcasting Optimal?

Proposition 1 (Optimality of broadcasting)

Broadcasting is optimal ($S^* = N$) if the interaction network satisfies at least one of the following conditions:

(i) **Bounded influence**: $d_i^{\text{in}} \leq 1 + d_i^{\text{out}}$ *for all* $i \in N$;

(ii) **Bounded centrality**: *for all* $i \in N$ *such that* $d_i^{\text{in}} > 1 + d_i^{\text{out}}$,

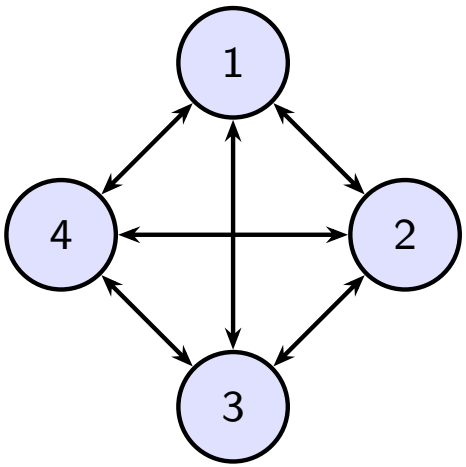
$$b_i^N \leq \frac{d_i^{\text{in}}}{d_i^{\text{in}} - d_i^{\text{out}} - 1};$$

(iii) **Uniform attentiveness**: $d_i^{\text{out}} = d_j^{\text{out}}$ *for all* $i, j \in N$.

- **"Less is more" requires asymmetric positions:** some agents must be *central* within the target set and *influential* in the full network.
- **"Less is more" requires imprecise communication:** as $\gamma \to 1$, y pins down θ and aligns everyone, so broadcasting is costless.

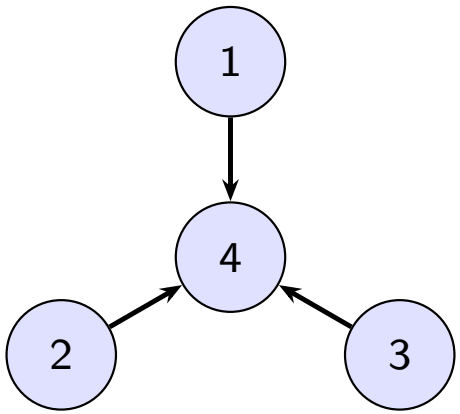
Where Broadcasting Is Optimal

- The following 4-player structures satisfy the conditions of Prop. 1 — so optimal scope is broadcasting ($S^* = N$):



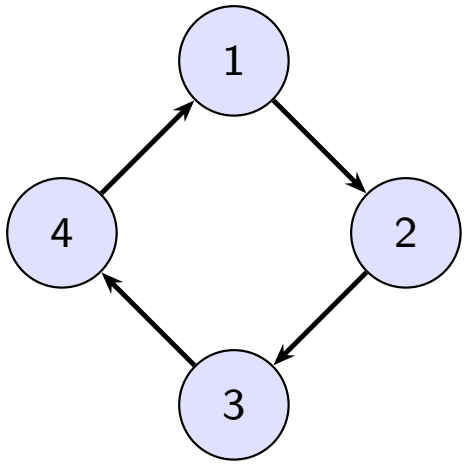
(A) Complete

($\Rightarrow$ i)



(B) Star

($\Rightarrow$ ii)

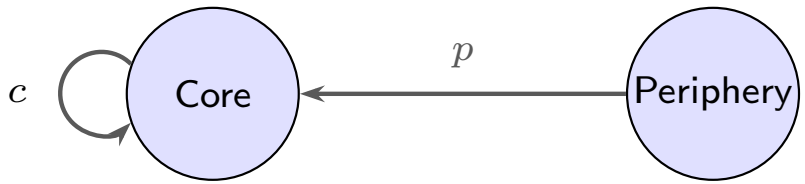


(C) Circle

($\Rightarrow$ iii)

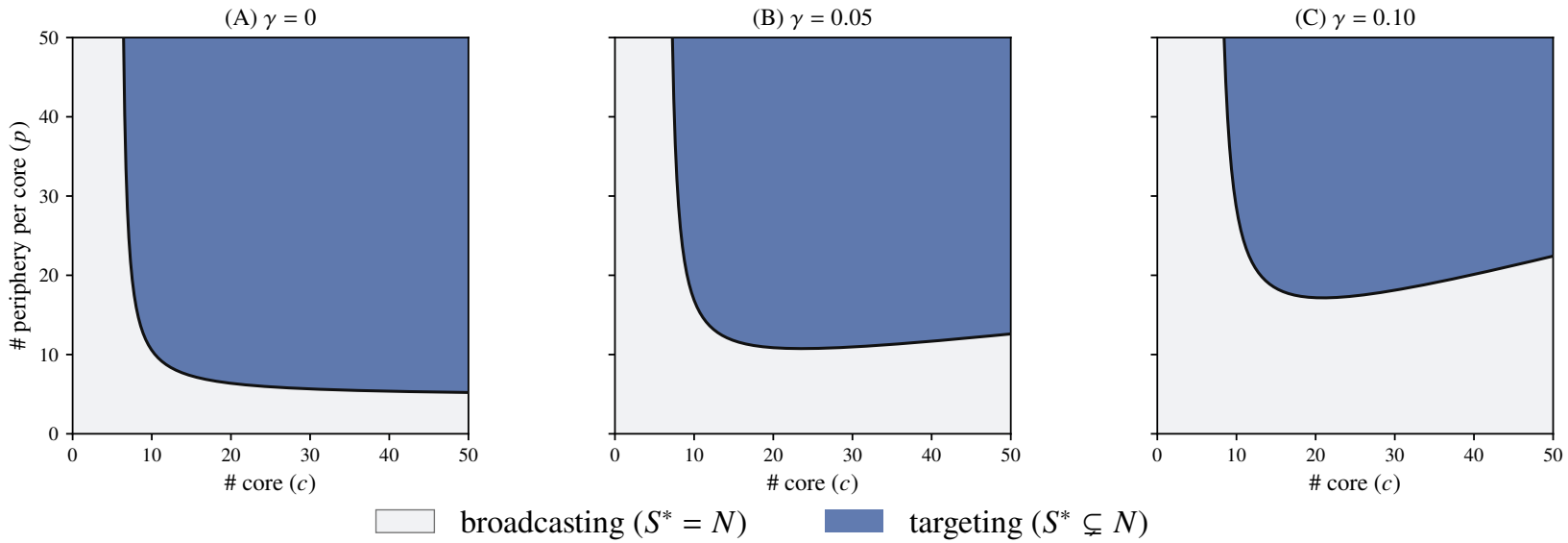
Core–Periphery: When Less is More

- A *complete-core* network with *private peripheries*:
  - **Core** (*KOLs*): every pair of core agents follows each other.
  - **Periphery** (*ordinary users*): periphery agent follows only its own core.



Note: nodes group agents by role. The loop c denotes a complete core of c mutually following agents; p denotes the number of private followers attached to each core agent, not a link weight. The arrow follows the listening convention $i \to j$.

Targeting Beats Broadcasting



Note: Links are binary $g_{ij} \in \{0, 1\}$.

- **Blue region:** $\mathcal{T}(N \setminus \{i\}) > \mathcal{T}(N)$ for $i \in$ Core, hence the optimal scope is strict: $S^* \subsetneq N$.
- **Comparative statics:** the targeting region contracts as the public signal becomes more precise (γ rises).

Optimal Targeting: Who Matters?

Optimal Targeting: The Key Recipient

- *If only a single agent can receive the public information, who should it be?*

Proposition 2 (Key recipient)

When the planner communicates with a single agent, the key recipient is any agent who maximizes:

$$D_i \equiv d_i^{\text{out}} + d_i^{\text{in}}, \tag{4}$$

which is the equally weighted sum of out- and in-degree. We refer to D_i as the **total degree** *measure of agent i.*

- **Intuition**: When only one agent is targeted, higher-order amplification disappears ($b_i = 1$). What remains are two first-order forces: the agent's attentiveness to others, captured by *out-degree*, and the agent's influence on others, captured by *in-degree*.

Optimal Targeting: The Key Group

- *If multiple agents can receive the public information, which set of recipients should the planner choose?*

Proposition 3 (Key group)

When the planner communicates with multiple agents, under sufficiently low transparency, finding the key group of recipients is **NP-hard** *in general.*

- **Intuition**: The planner prefers recipients who are *influential* but *not* tightly linked to one another. Selecting the largest such set of mutually-unlinked hubs is related to **maximum independent set (MIS)**.
- **Remark**: No polynomial-time algorithm can be guaranteed to recover the exact key group for every network, however, approximation methods such as *greedy* algorithms can be useful.

Conclusion

- **Model:** We study a network game in which agents coordinate under dispersed information and a benevolent planner chooses the scope of public communication.

- **Main message:** Broadcasting is often optimal, but targeting can improve welfare when influential and central recipients excessively amplify noisy public information.

Thank you. Comments very welcome.